ICT CORNER

Dual nature of radiation and matter

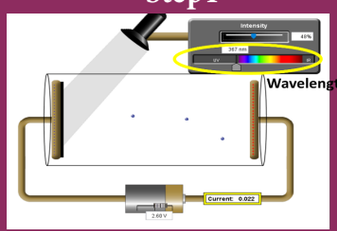
In this activity you will be able to visualize how light knocks electrons off a metal target and describe the photoelectric effect experiment.

Topic: Photoelectric effect

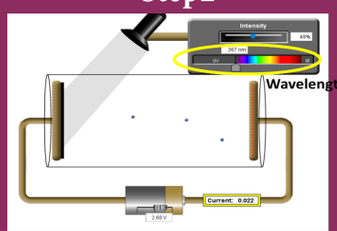
STEPS:

- Open the browser and type "https://phet.colorado.edu/en/simulation/legacy/photoelectric" in the address bar.
- Change intensity of light and observe how the intensity of light will affect the photo electric current and the energy of electrons
- By adjusting the value of wavelength and observe how the wavelength of light will affect the photo electric current and the energy of electrons
- Adjust the value of voltage from the battery and analyse the effect of potential difference on the photoelectric current.
- Change the material of the target and analyse how it will affect the current and the energy of electrons.
- Study the photo electric current – voltage graph and Photo electric current - intensity graph obtained from this experiment.

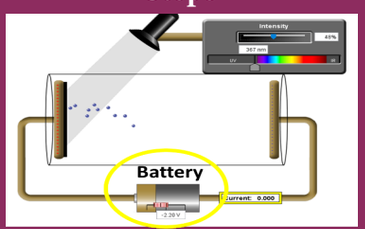
Step1



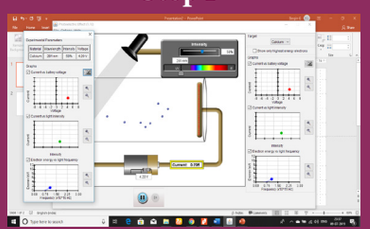
Step2



Step3



Step4



Note:

Install Java application if it is not in your browser.

You can download all the phet simulation and works in off line fromhttps://phet.colorado.edu/en/offline-access .

URL:

https://phet.colorado.edu/en/simulation/legacy/photoelectric

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* Pictures are indicative only.

* If browser requires, allow **Flash Player** or **Java Script** to load the page.

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